# VU HOANG LONG

**Senior Software Engineer**

**Email**: longhv.dev@gmail.com **Location**: Ho Chi Minh City

---

## Professional Summary

**Senior Software Engineer** specializing in **backend architecture**, **event-driven systems**, and **multi-tenant platform design**. Proven ability to **deliver reliable services, simplify complex integrations, and maintain high-quality codebases**. Focused on **building practical systems that scale with strong operational discipline**.

---

## Skills

- **Back-End Development**: **TypeScript**, **NestJS**, **PostgreSQL**, **SQS**, **EventBridge**, **Lambda**, HTTP API, WebSocket, SSE.
- **Front-End Development**: **React**, react-native, **MobX**, **zustand**, Ant Design, MUI, TailwindCSS, HTTP API, WebSocket, SSE, react-router, react-query, react-hook-form, react-spring, usehooks.
- **Others**: **Git**, GitLab CI, **Docker**, Cloudflare, Linux.

---

## Professional Experience

**Software Engineer** @ XOLV Technology Solutions (09/2025–08/2026)

**Project Objective**: A healthtech platform focused on improving the lives of individuals impacted by ASD/IDD and supporting the caregivers around them.

**Key Technologies**: TypeScript, NestJS, PostgreSQL, SQS, EventBridge, Lambda.

**Clinical Patient Onboarding System**

**My Role**:

- Implemented an end-to-end patient onboarding process from external surveys into the internal system using an **event-driven architecture**.
- Built a proactive client and appointment syncing workflow to keep data aligned across systems.
- Applied an **AI-driven agile development process** with the **BMAD-method** framework to accelerate delivery while maintaining a high-quality codebase.

**Multi-Tenancy Initiative**

**My Role**:

- Drove the initiative to officially support multi-tenancy for a legacy system.
- Migrated from application-level isolation to database-level isolation with **PostgreSQL RLS**, replacing multi-hop verification with **0-hop verification** without changing business code.
- Replaced the legacy unstandardized JWT self-issuance flow with a secure and simple tenant onboarding process using **AWS Cognito OAuth 2.0 IDP**.
- Implemented a low-overhead and reliable database-level audit trail based on **WAL**.

**Platform Enablement**

**My Role**:

- Added the foundation for **feature flags** to support a **trunk-based development** model and enable a fast, non-blocking release process.
- Improved observability by standardizing logging to the **OTel** standard and enabling end-to-end traceability.

**Software Engineer** @ Confidential Company (01/2019–05/2025)

**Project Objective**: Apply **microservices** to build a **scalable and high-performance online gaming platform**, integrating third-party games, supporting online payments, and providing **stable operations for a large user base**.

**Back-End Development** (primary focus)

**Technologies Used**: TypeScript, NestJS, PostgreSQL, Redis, Kafka, RabbitMQ, Docker.

**My Role**:

- Collaborated with stakeholders to define business requirements and translate them into **robust backend API designs** and **database schemas**.
- Designed and implemented a **DDD high-load microservices** backend architecture to **handle large volumes of concurrent users**.
- Created multi-stage Docker images for various services to support CI/CD workflows.
- Monitored backend performance; optimized query speed through PostgreSQL indexing and implemented batch inserts to improve write throughput.
- Applied backend performance enhancements; implemented rate limiting and Cloudflare protections to mitigate bot traffic and prevent DDoS attacks.
- Participated in incident response and bug fixing during live operations.

**Front-End Development**

**Technologies Used**: TypeScript, ReactJS (with Vite), React Router, React Hook Form, HTTP API, WebSocket, SSE, zustand, TailwindCSS, react-spring, usehooks.

**My Role**:

- Integrated UI components with **backend APIs**, ensuring seamless data flow and full application functionality.
- Implemented real-time features using **WebSocket and Server-Sent Events (SSE)**, significantly enhancing user engagement and responsiveness.
- Developed a highly **responsive** and navigable user interface using ReactJS and react-router, styled with TailwindCSS for consistent and adaptable layouts across various screen sizes.
- Streamlined form handling and improved data validation through the implementation of react-hook-form, leading to a more efficient and error-free user input process.
- **Managed application state** effectively using zustand, contributing to improved performance and simplified component logic.
- Enhanced user experience through the integration of smooth and engaging animations using react-spring.

**Key Outcomes & Achievements**:

- Successfully built and operated a scalable online gaming platform with **80,000 active users** and a peak **CCU of ~1,500**, capable of handling up to **6,000 requests per second**.
- Achieved **95% of requests responding under 200ms** through backend performance optimizations such as PostgreSQL indexing, batch inserts, and application-level as well as Cloudflare rate limiting to mitigate bot traffic and DDoS attacks.

- Delivered a **secure, responsive, and user-friendly UI/UX** with **real-time updates** powered by **WebSocket and SSE**, ensuring seamless integration with backend systems.
- Enhanced user engagement and responsiveness through smooth animations, efficient form handling, and optimized application state management.
- Provided a fully integrated and interactive cross-device experience, combining high-performance backend architecture with a modern, intuitive front-end.

---

# Education

**Bachelor of Software Engineering** @ Thai Nguyen University (2018–2021)

---

# Languages

**Vietnamese**: Native

**English**: Professional